DATA 3464: Fundamentals of Data Processing

Wrangling text

Charlotte Curtis

March 5, 2026

Topic overview

- Text file formats and character encodings
- Extracting features from (un)structured text
- Regular expressions

Resources used:

- Python documentation
- Joel on Software Blog Post

What is a text file?

- Any file is just a sequence of **bytes**
- The file **extension** is somewhat meaningless
- Text files contain only human-readable **characters**
- **Binary files** are everything else, including:
    - Images
    - PDFs
    - Word documents*
    - Executables

Technically word docs are just zip files with XML (structured text) inside

Character encodings

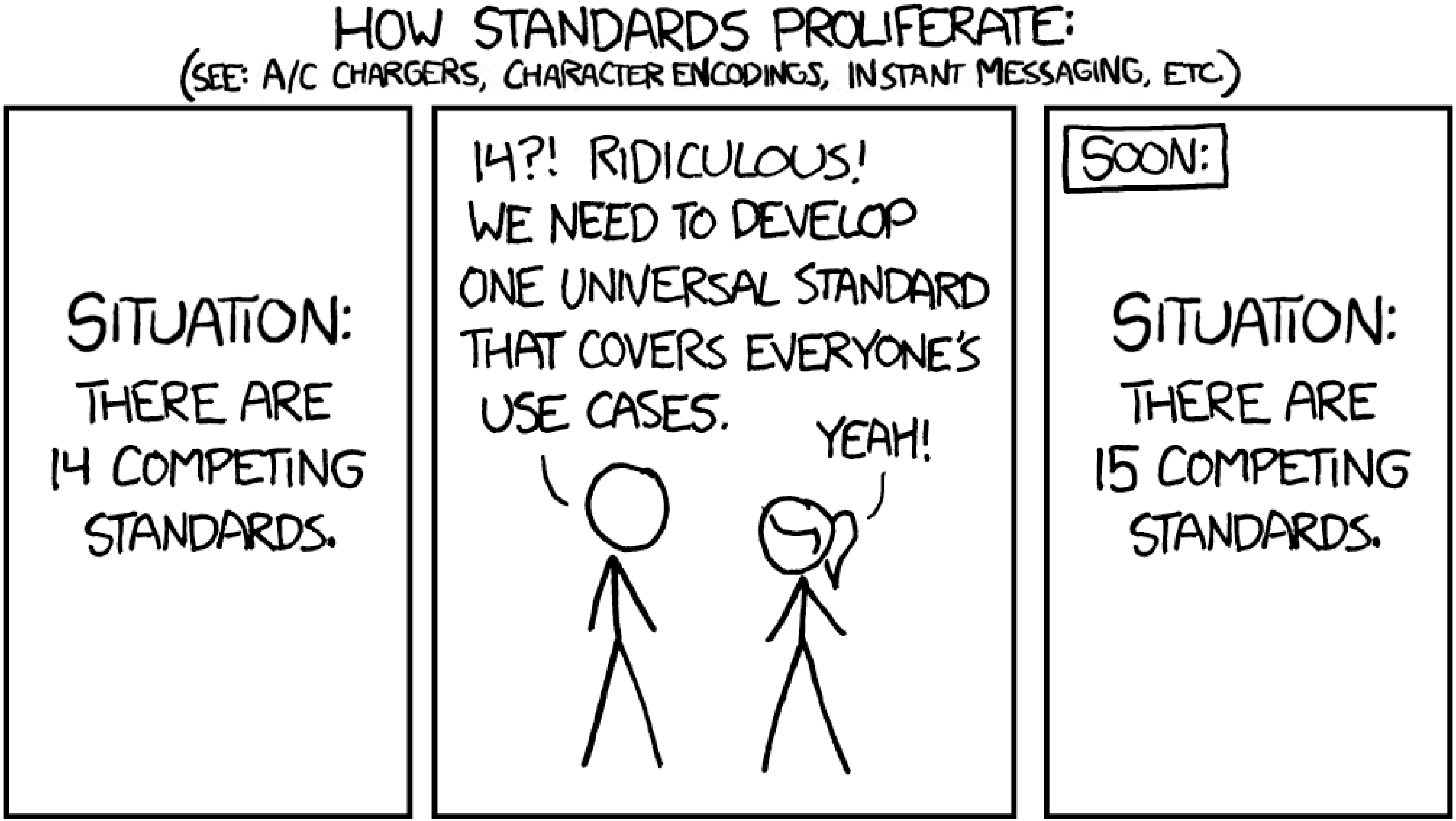
> There are three hard problems in computer science:
>
> 1) Converting from PDF,
>
> 2) Converting to PDF, and
>
> 3) O3[illegible]
>
> —Marseille Folog

- To properly read a text file, we need to know its **encoding**
- Character encodings define how bytes are interpreted
- Turns out this is surprisingly complicated

ASCII

- American Standard Code for Information Interchange (1963)
- 7-bit encoding (128 characters)
- First 32 are **control characters** (e.g., newline, tab)
- Then punctuation, digits, uppercase letters, lowercase letters
- Most computers use 8-bit bytes, so there's a whole 128 characters "left over"

> *This is a very English-centric character set!*



https://xkcd.com/927/

Unicode to the rescue

- In the 1980s things were already getting out of hand
- The **Unicode consortium** published a standard in 1993 that assigned a **code point** to every character they could think of (297,334 as of Unicode 17.0)
- Currently, the most common encoding is UTF-8
  - "Unicode Transformation Format – 8-bit"
  - How can 297k characters be represented in only 8 bits?
- The first 128 characters are 1 byte each and align with ASCII
- After that, 2-4 bytes per character are used

> *There's also UTF-16 and UTF-32, but now we need to deal with endianness*

Line endings

- Say we agree to use UTF-8 👍
- A text file is still just a big long string of **bytes**
- Humans like things to be orderly and readable with **line breaks**

System	Abbreviation	Escape sequence	Code point
Pre-OS X Mac	CR (carriage return)	`\r`	U+000D
Unix	LF (line feed)	`\n`	U+000A
Windows	CRLF	`\r\n`	U+000D U+000A

- Result: chaos, but we've mostly settled on LF (`\n`)

Why am I telling you all this?

- As data scientists, you will need to ingest data from various sources
- You **will** encounter character encoding and/or line ending issues
- You don't need to memorize all this, but recognizing that an issue exists will go a long ways towards fixing it

> *Example: misadventures with "smart" quotes*

Where we left off on March 5

Portable document format

- The PDF specification was first published in 1993
- PDFs are like a "digital paper" that can contain text, images, vector graphics, and more (like JavaScript, forms, and weirdly, 3D models)
- We use them for all sorts of things because the **appearance** is consistent
- They are absolutely terrible for pretty much everything else

Some useful PDF packages:

- pdfminer.six: extract text and metadata
- pdfplumber: built on pdfminer.six, layout aware text extraction
- pikepdf: low level PDF manipulation
- pymupdf: more low level PDF manipulation
- and many (many) more!

> *DATA 3463 covers more about PDFs, including OCR. The main focus in this class is on fixing the issues and dealing with edge case*

Hypertext markup language

- Much simpler than PDFs, but still not great for data exchange
- Markup languages are structured text files that define how content *should* be displayed, but it's not always consistent
- Most of the internet is UTF-8 encoded HTML
- We can try to extract text with something like BeautifulSoup
    - Again, more on web scraping in DATA 3463

Back to text

- Assuming we've got text from somewhere, we probably want to:
    - Organize it into a structured format (csv, database, json)
    - Identify specific features (postal codes, dates, names)
- Need to deal with encoding issues, garbled sentences, mixed up tables, and all the other bizarre ways things go wrong
- There is no magic flowchart for this! Lots of trying things, seeing what happens, dealing with a few edge cases at a time

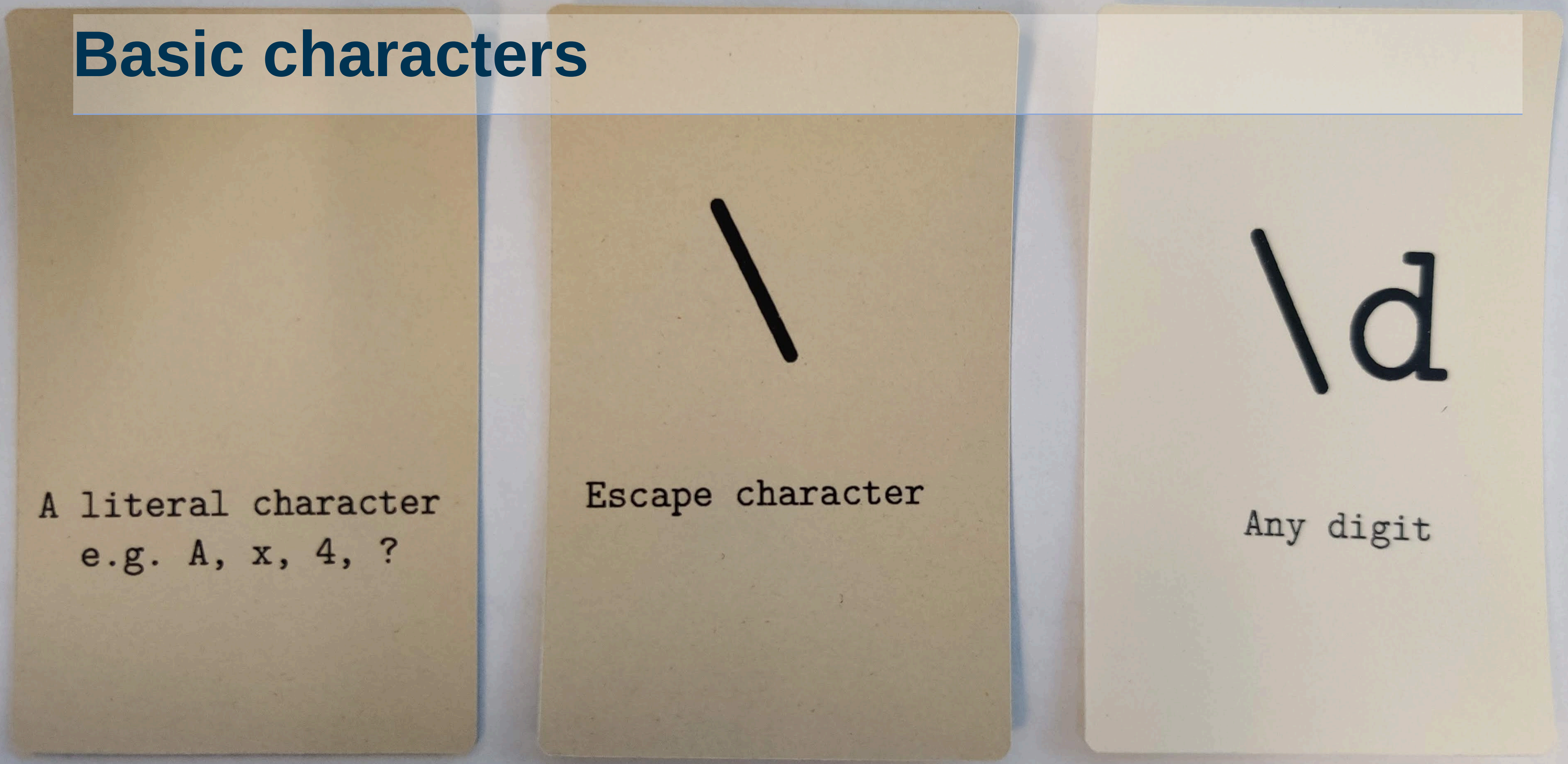
> *One really useful tool is **regular expressions*** 🔨

Regular expressions

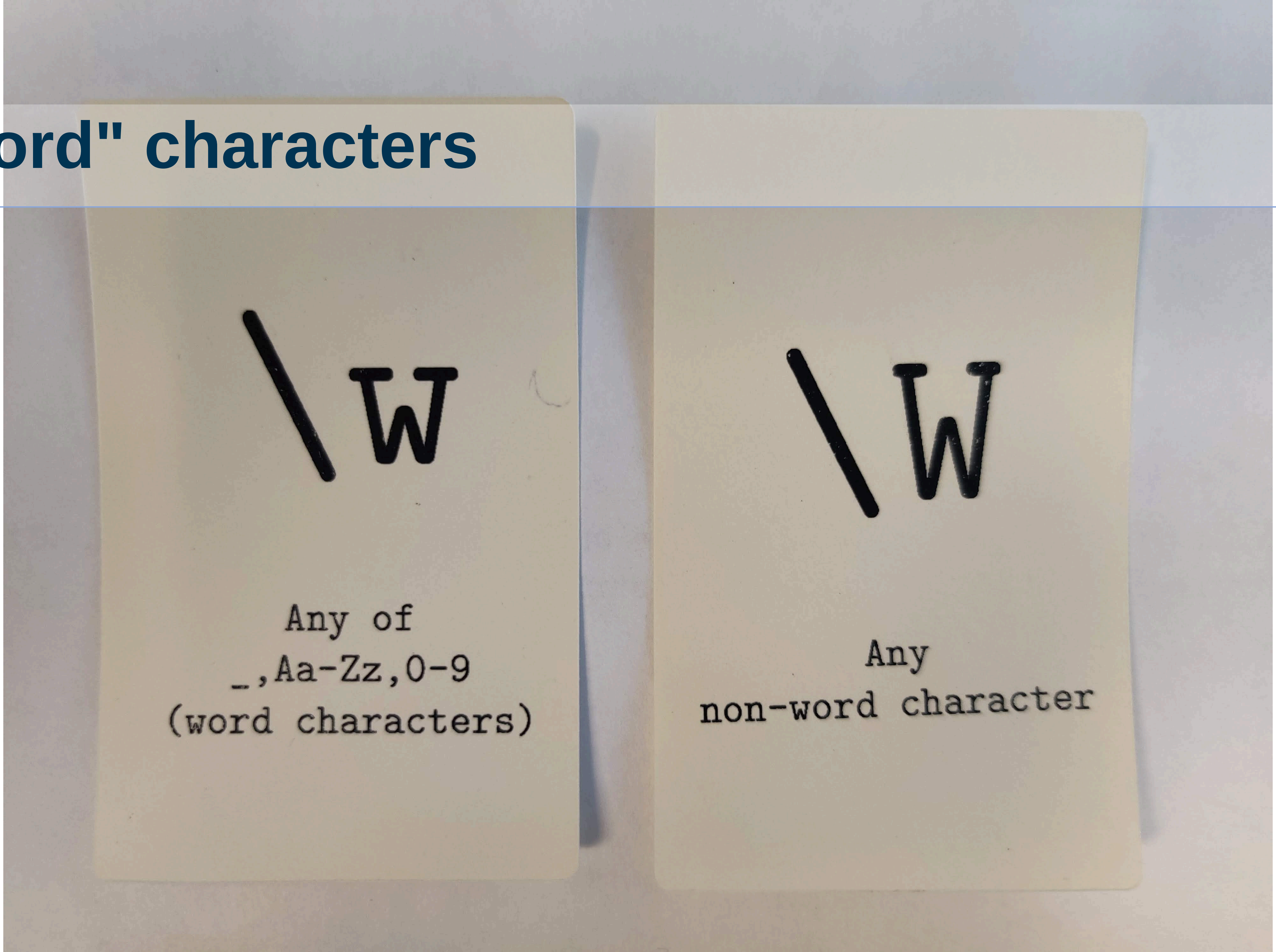
The card game?

Note: card groupings and colours are logical where possible, but sometimes just random

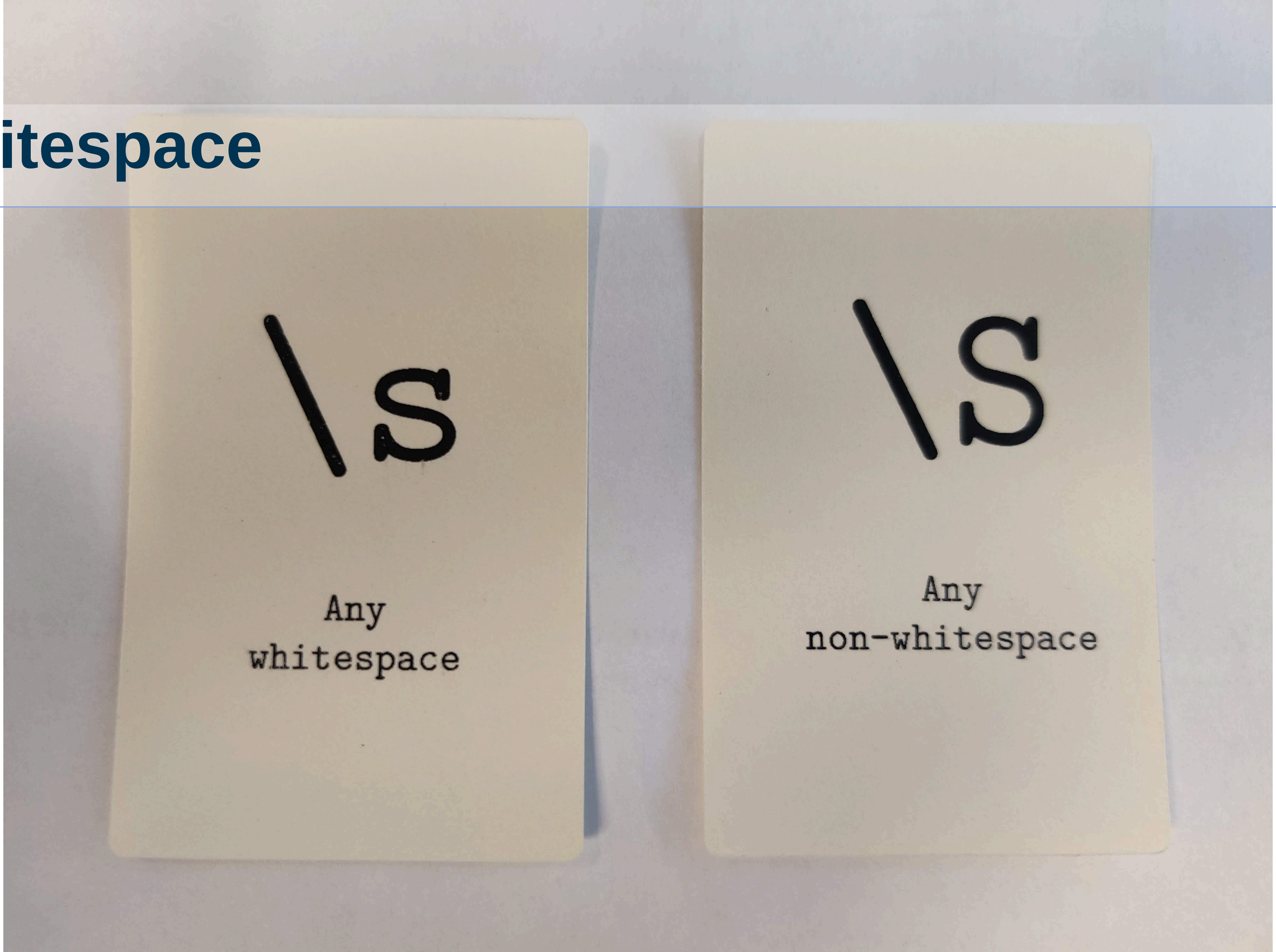
Basic characters



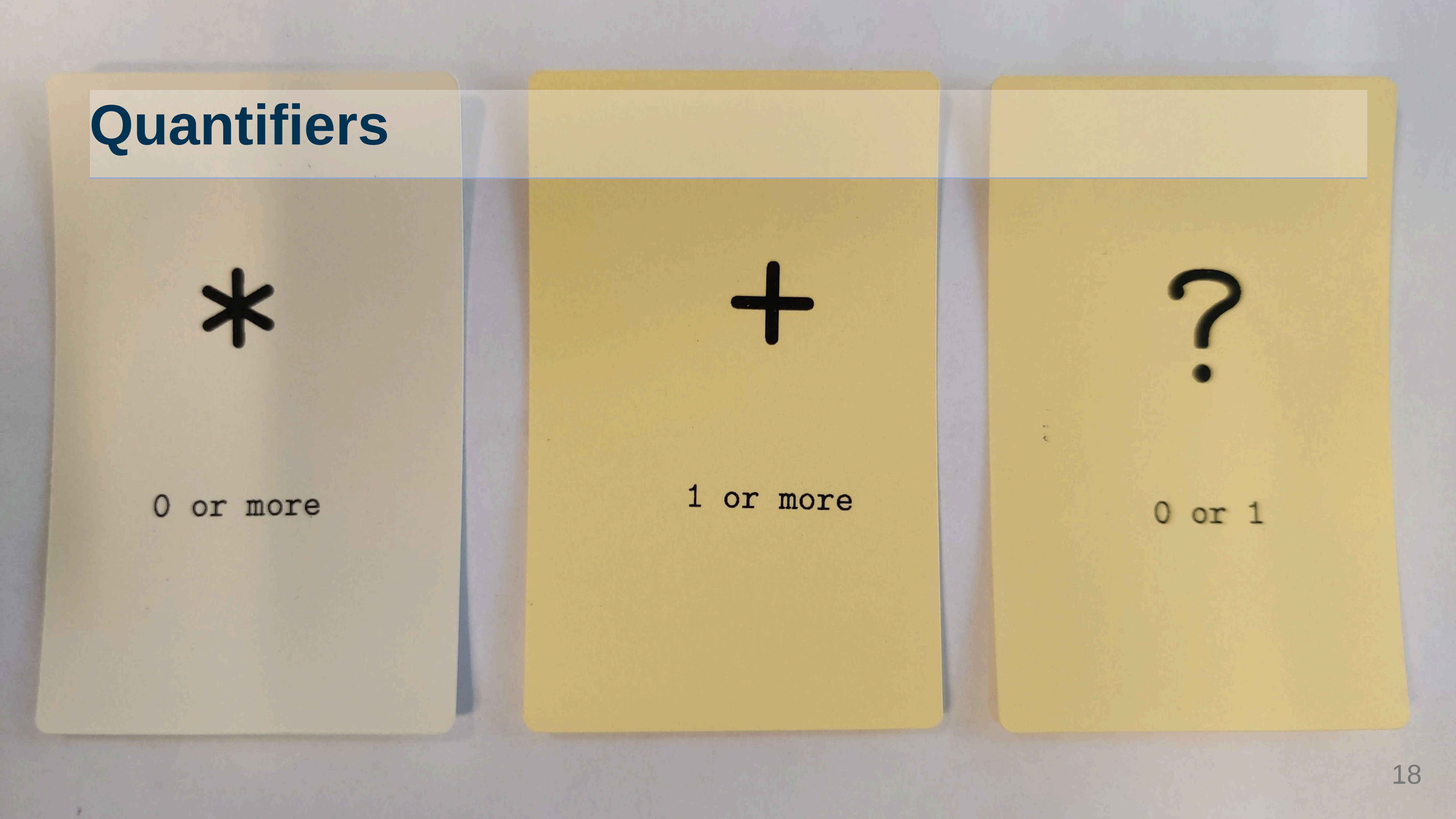
"Word" characters



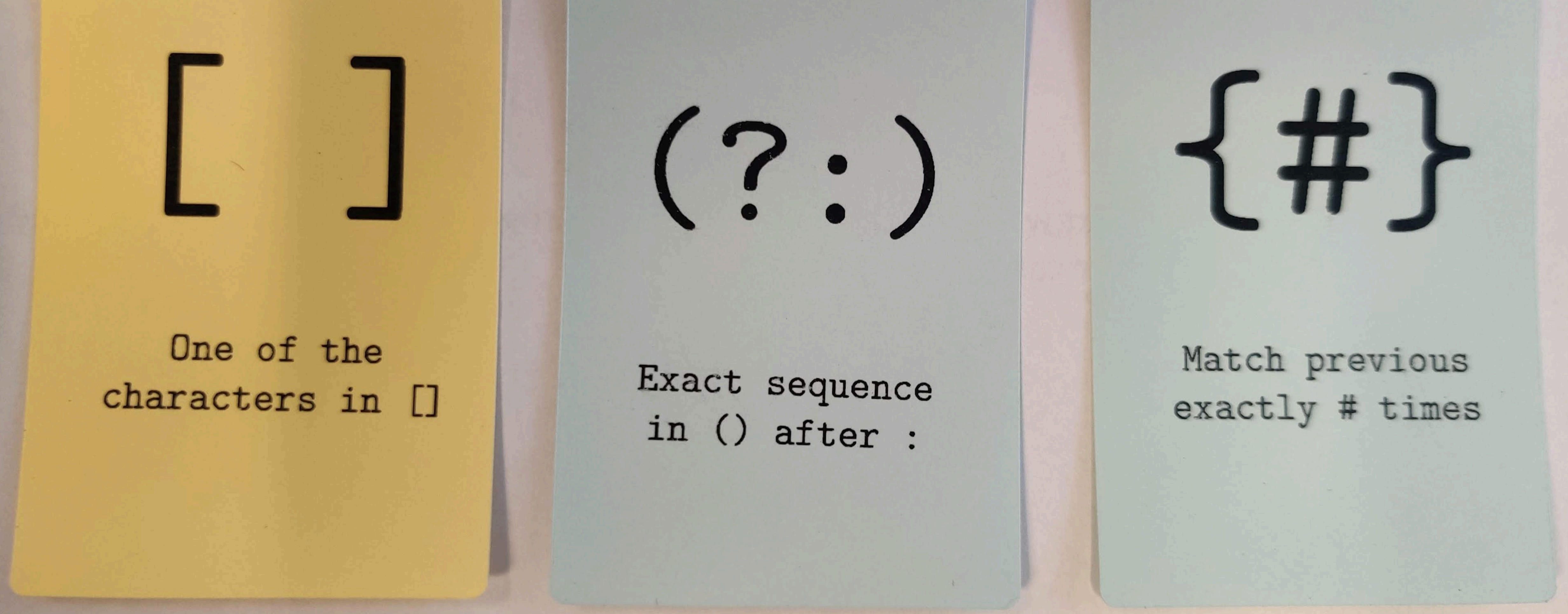
Whitespace



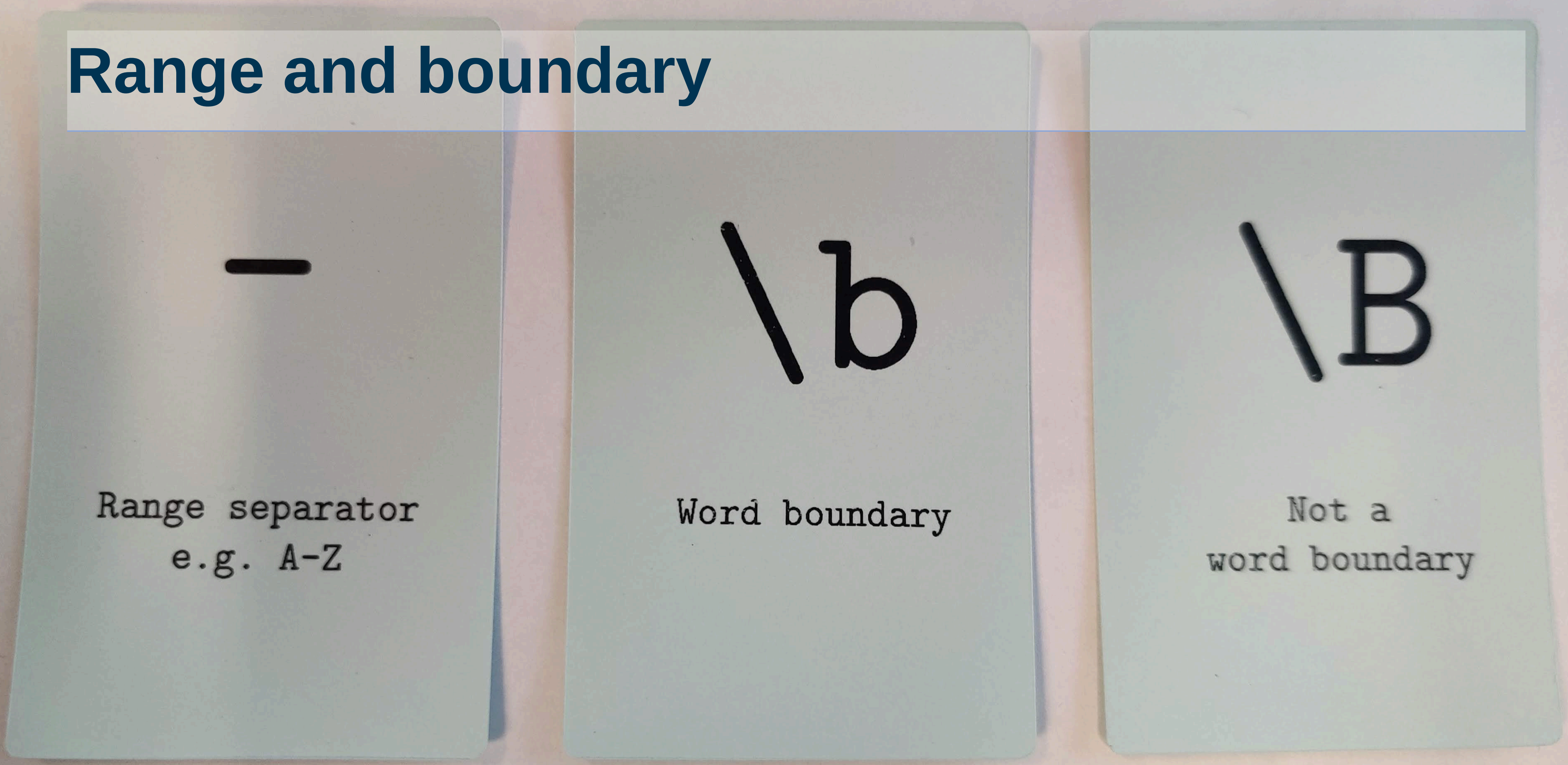
Quantifiers



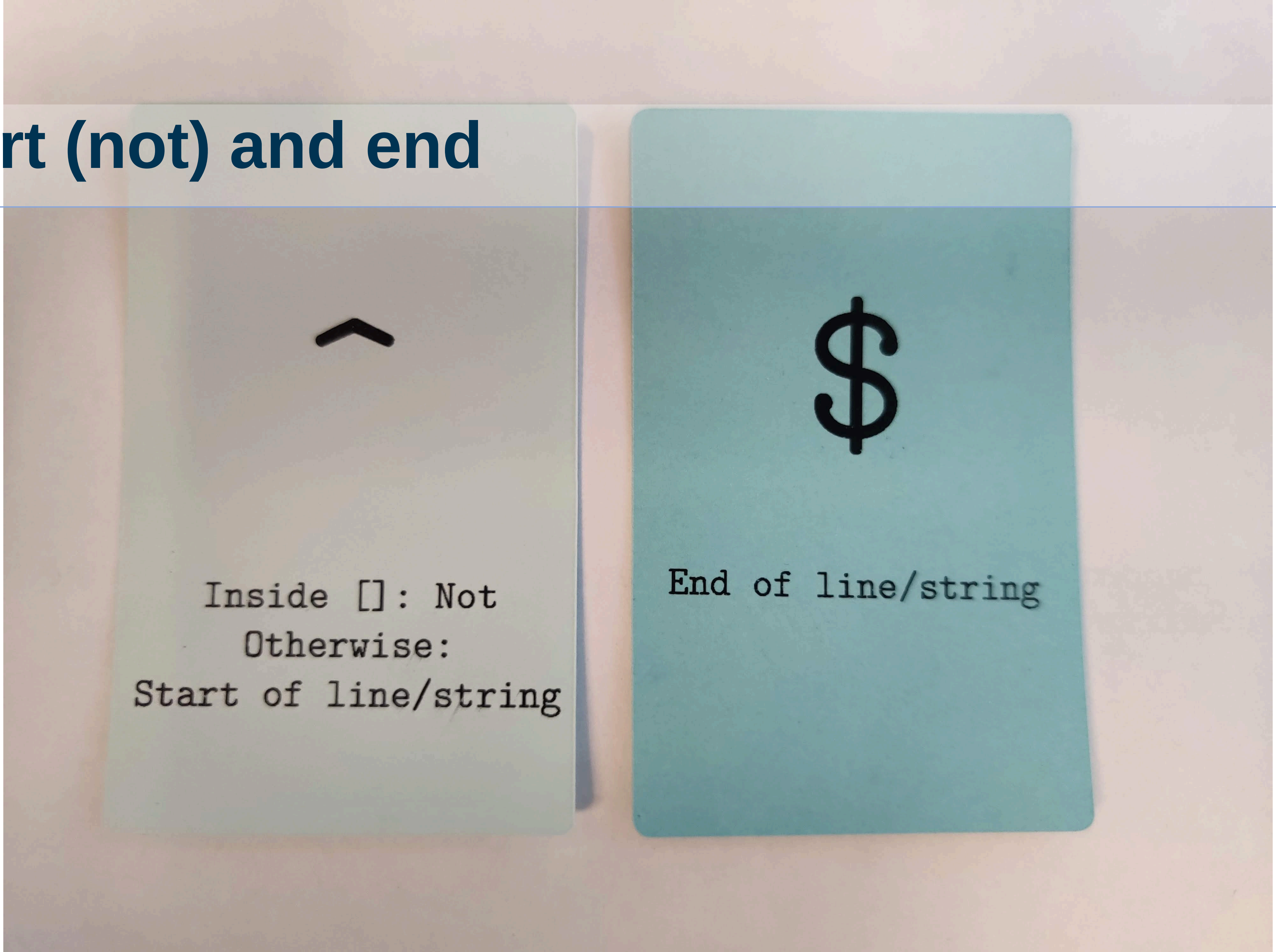
Brackets and braces



Range and boundary



Start (not) and end



Where we left off on March 10

Case study: Cloudflare outage

In July 2019, a poorly formed regex in Cloudflare's firewall rules caused CPU usage to spike and websites to come crashing down worldwide. The cause? This regex:

```
(?:(?:\"|'|\]|\}|\\|\d|(?:nan|infinity|true|false|null|undefined|symbol|math)|\`|\-|\+)+[)]*;?((?:\s|-|~|!|{}|\|\||\+)*.*(?:.*=.*)))
```

- Regex engines perform **backtracking** to check for multiple match possibilities
- This particular string led to **catastrophic backtracking**
- Moral of the story: test carefully, both positive and negative

> *I like https://regex101.com/ for quickly testing regular expressions*

Coming up next

- Assignment 3: curate a dataset
- A primer on data cards
- Intro to signals and images